and the result would be that after a certain length of time, the electron would have to fall into the nucleus and come to rest there. This picture is totally different from the view presented by the atom model with its electrons circling in their orbits undisturbed. And even if it were possible in some way or other to get over this question of radiation, Rutherford's atom model would still fail entirely to account for the stability and definiteness of the chemical properties of the atom.

The problem of co-ordinating atomic stability with Rutherford's atom model was solved by Niels Bohr in 1913. Bohr's views were based on a theory of Max Planck. In 1900, Planck had formulated—at first purely empirically—his radiation law, which interpreted the thermal radiation of a 'black body' (a body which absorbs all the radiation incident on it, and consequently—according to a law formulated by Kirchhoff—is also the most powerful emittor of radiation) in conformity with actual experience. In his subsequent attempt to give this law a physical interpretation, Planck encountered a very peculiar discontinuity in natural processes. He found that he was able to establish his radiation law solely on the foundation of the remarkable hypothesis that the smallest radiating particles, the atoms, could not assume the continuous sequence of all possible energy values of their vibrations (as they would have been expected to, according to all previous knowledge) but only a series of certain definite, specific energy values. It even appeared—and later research proved it, too—that the emitted radiation manifested this quality of discontinuity, and that light, regarded up to then as a wave process, must also consist of discrete quanta of energy. Planck realized that light of the frequency ν must be emitted as well as absorbed in discrete quanta of energy, of a magnitude proportional to the frequency and he postulated that the energy of these quanta was equal to $h\nu$. This constant, h, is called *Planck's constant* (see Table 1), and has been the keynote of the entire development of physics for several decades.

As a result of this theory of energy quanta of light, a very strange situation ensued. On the one hand, it was recognized clearly that certain optical phenomena—e.g. those of interference—could only be understood by regarding light as a wave process. On the other hand, the concept of light quanta appeared to be no less necessary in order to explain other phenomena. But according to the latter view, light has no longer the characteristics of waves, propagated in space in all directions, but is a swarm of particles traversing space in straight lines. Therefore, it appears to be impossible to do without two completely different and fundamentally contradictory views of light. We can thus speak of a *wave-particle duality*—of a wave aspect and a corpuscular or particle aspect of light.

Bohr started out from the above-mentioned hypothesis of a discontinuity in atomic phenomena, and he advanced the theory that atoms can exist for an